Quantification: 548_hyp_HP_z14c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	24
interior lacuna count	16
frame-edge lacuna count	8

Interior lacunae only: mean, median and sample SD (n = 16)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2916.25	1753	2601.37	px^2
major_axis_length_px	lacuna shape	83.1915	72.0264	39.7271	px
minor_axis_length_px	lacuna shape	44.262	38.318	20.2646	px
aspect_ratio	lacuna shape	1.9146	2.009	0.4049	unitless
eccentricity	lacuna shape	0.8207	0.8673	0.1046	unitless
solidity	lacuna shape	0.7857	0.7974	0.074	unitless
roots_count	headline	9.6875	7	6.183	count
ring_length_r30_px	headline	310.375	257.5	193.492	px
ring_length_r60_px	reported	619.5	568	364.964	px
owned_length_px	ownership-dependent	1017.79	775.175	869.752	px
edge_count	ownership-dependent	53.1875	35.5	47.1568	count
mean_edge_length_px	ownership-dependent	19.8908	18.7241	4.6954	px
perimeter_px	normalised	293.56	221.622	177.978	px
ring_area_r30_px2	normalised	8852.06	7623	4532.65	px^2
ring_area_r60_px2	normalised	18871.5	18514.5	9034.8	px^2
in_frame_fraction_r30	normalised	0.9679	1	0.0681	unitless
in_frame_fraction_r60	normalised	0.9409	1	0.1121	unitless
ring_density_r30	normalised	0.0342	0.0335598	0.00743478	px^-1
ring_density_r60	normalised	0.0318123	0.0321693	0.00581292	px^-1
roots_per_100px_perimeter	normalised	3.4619	3.1219	1.3861	per 100 px
ring_attached_length_r30_px	network v2	280.438	221.5	187.668	px
ring_attached_length_r60_px	network v2	495.25	395	340.62	px
ring_length_w_r30_px	network v2	354.82	297.179	220.691	px
ring_length_w_r60_px	network v2	707.002	643.701	418.2	px
sholl_crossings_r10	network v2	7.3125	5.5	4.1748	count
sholl_crossings_r20	network v2	8.125	7.5	5.201	count
sholl_crossings_r30	network v2	8.1875	6.5	5.5644	count
loop_count	loops, report only	3.375	2.5	3.4617	count
loop_edge_share	loops, report only	0.1879	0.2375	0.1283	share
ring_width_mean_r30_px	width	7.703	7.5733	0.5866	px

Per field
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measure	value	units
analysed_area_px2	980525	px^2
total_skeleton_length_px	27510	px
canalicular_length_density_per_px	0.0280564	px^-1
junction_count	594	count
junction_density_per_px2	0.0006058	px^-2
lacuna_count	24	count

lacunae_per_px2	2.448e-05	px^-2
skeleton_component_count	380	count
mean_component_length_px	72.3947	px
median_component_length_px	19	px
field_length_density_w_per_px	0.0319453	px^-1
field_density_without_flagged_per_px	0.0280564	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	248	px
unattached_component_count	14	count
canalicular_length_density_attached_per_px	0.0280564	px^-1
field_length_density_with_unattached_per_px	0.0283093	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	76	count
loop_edge_share	0.1956	share
edge_count	1539	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.8711	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	27510	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2147	91.4001	39.1166	2.3366	0.9038	0.7373	0.5027	31.0005	542	True
2	2407	118.141	31.4268	3.7592	0.964	0.7314	0.0789	45.1836	688.173	True
3	704	43.5023	24.5984	1.7685	0.8248	0.8253	0.3934	22.6634	726.641	True
4	3074	97.1472	47.5979	2.041	0.8717	0.7951	0.7084	40.3224	458.502	False
5	4688	114.763	56.7615	2.0218	0.8691	0.8409	-0.5667	73.3957	882.268	False
6	996	48.8421	30.1974	1.6174	0.786	0.83	-0.3922	177.571	118.507	False
7	1527	72.7312	32.1897	2.2595	0.8967	0.7689	1.1031	215.125	119.538	False
8	791	45.4603	27.379	1.6604	0.7983	0.777	-1.1935	276.899	15.5398	True
9	885	49.967	25.4783	1.9612	0.8602	0.8157	0.1758	321.858	27.7831	False
10	865	60.9988	23.7735	2.5658	0.9209	0.67	-0.6206	306.805	864.442	False
11	9764	174.84	79.9051	2.1881	0.8895	0.756	-1.0035	341.592	370.654	False
12	5590	100.499	74.9477	1.3409	0.6662	0.8931	0.3458	390.83	706.214	False
13	772	53.4057	21.7698	2.4532	0.9131	0.7546	-1.3439	371.161	29.9093	False
14	3806	88.412	63.5037	1.3922	0.6958	0.7996	0.5855	456.034	447.32	False
15	1368	68.8013	32.6693	2.106	0.8801	0.7621	0.1327	437.326	13.0395	True
16	818	48.7035	23.1908	2.1001	0.8794	0.8503	0.5594	490.792	650.099	False
17	583	30.9882	25.7454	1.2036	0.5565	0.8807	-0.6814	486.81	163.602	False
18	6173	146.612	73.4479	1.9961	0.8655	0.639	0.4437	581.134	750.702	False
19	1268	55.1715	37.4316	1.4739	0.7346	0.792	0.9741	664.745	233.587	False
20	1408	71.6194	28.9459	2.4743	0.9147	0.8167	1.0982	723.629	987.69	True
21	3872	116.659	53.0473	2.1992	0.8906	0.6702	-1.541	793.571	878.724	False
22	3821	95.6808	52.75	1.8139	0.8343	0.9065	-1.1603	884.495	983.684	True

23	1979	71.3216	39.2045	1.8192	0.8354	0.8154	-1.307	916.707	906.786	False
24	8745	138.777	89.9937	1.5421	0.7612	0.8115	1.2047	976.542	708.753	True

Network per lacuna

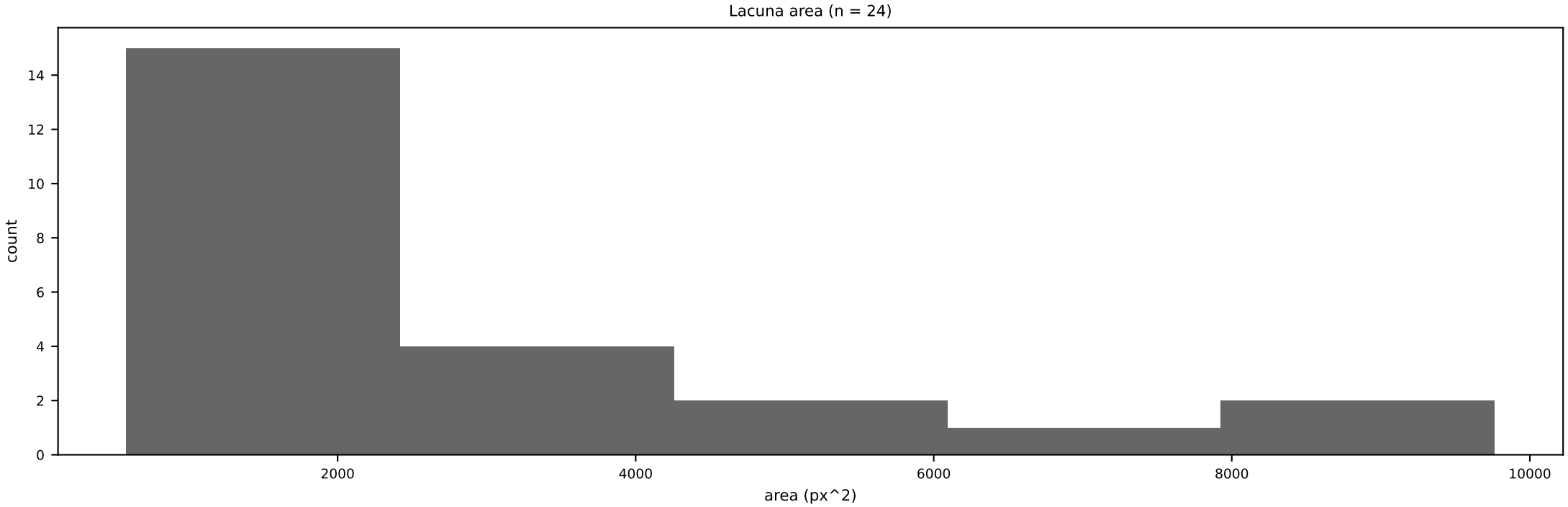
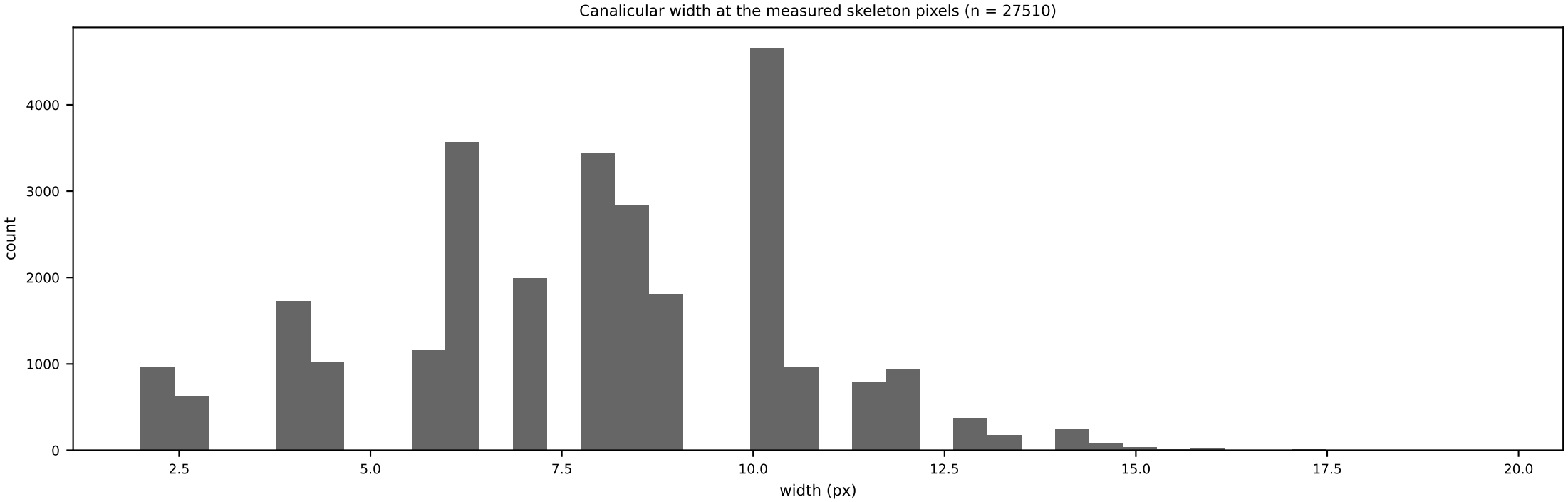
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	7	191	343	781.281	31	25.2026	6.9981
2	11	324	632	590.357	36	16.3988	8.0085
3	4	141	284	467.961	23	20.3461	5.7534
4	10	241	509	1030.68	59	17.4691	7.7676
5	7	288	588	736.566	22	33.4803	8.7175
6	4	123	248	551.132	27	20.4123	8.0682
7	6	172	421	613.044	22	27.8656	7.5689
8	3	151	227	63.1838	2	31.5919	5.7979
9	6	70	94	75.9411	4	18.9853	6.6883
10	12	321	689	1428.11	80	17.8514	7.3687
11	24	778	1522	3355.14	183	18.3341	7.3534
12	9	469	961	2214.75	109	20.3188	8.7134
13	5	168	247	225.652	12	18.8043	7.2723
14	10	353	751	950.838	51	18.6439	7.1225
15	1	59	133	83.8406	4	20.9602	7.0119
16	4	218	399	463.345	22	21.0611	7.4299
17	7	184	566	283.451	21	13.4977	7.9669
18	22	607	1128	1944.31	98	19.8399	7.5777
19	7	274	570	813.784	44	18.4951	7.2242
20	8	311	574	1424.43	70	20.3489	7.755
21	17	525	861	1278.74	78	16.3942	7.831
22	6	201	375	404.09	23	17.5691	8.0704
23	5	175	358	319.208	19	16.8004	8.5769
24	7	428	897	763.896	40	19.0974	6.8574

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	314.877	6967	12578	0.7247	0.6734	0.027415	0.0272698	2.2231
2	330.693	7406	15690	0.7916	0.7301	0.0437483	0.0402804	3.3263
3	132.368	3187	6317	0.7504	0.6702	0.0442422	0.044958	3.0219
4	289.563	8243	15900	0.836	0.748	0.0292369	0.0320126	3.4535
5	351.161	12196	26476	0.9903	0.8855	0.0236143	0.0222088	1.9934
6	192.989	4006	9474	1	1	0.0307039	0.0261769	2.0727
7	221.38	6390	14742	1	1	0.0269171	0.0285579	2.7103
8	143.74	3169	5742	0.6661	0.6142	0.0476491	0.0395333	2.0871
9	148.61	2877	4028	0.8558	0.7127	0.0243309	0.0233366	4.0374
10	184.652	7467	20230	1	1	0.0429891	0.0340583	6.4987
11	750.649	18042	35989	1	1	0.0431216	0.0422907	3.1972
12	352.919	11920	25165	1	1	0.0393456	0.038188	2.5502
13	186.959	3412	6249	0.8041	0.7079	0.049238	0.0395263	2.6744
14	328.233	11363	23232	1	1	0.0310657	0.0323261	3.0466
15	193.539	4187	8710	0.6366	0.5949	0.0140912	0.0152698	0.5167
16	137.196	6367	12630	1	1	0.0342391	0.0315915	2.9155
17	104.569	5596	16799	1	1	0.0328806	0.0336925	6.6942
18	631.595	16663	33826	1	1	0.036428	0.0333471	3.4832
19	199.865	7779	21047	1	1	0.035223	0.0270822	3.5024
20	197.48	6605	13178	0.8037	0.7245	0.0470855	0.0435574	4.051
21	394.747	12775	22894	1	1	0.0410959	0.0376081	4.3066
22	281.38	6112	10015	0.7455	0.6918	0.0328861	0.0374438	2.1323
23	221.865	6537	13263	1	1	0.0267707	0.0269924	2.2536

24 525.044 12017 25181 0.7555 0.698 0.0356162 0.0356221 1.3332

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	181	317	208.64	386.563	4	3	5
2	259	372	377.848	734.24	5	7	10
3	122	246	165.196	327.664	4	3	3
4	188	413	275.451	575.931	5	7	4
5	278	481	336.363	685.21	9	9	10
6	123	182	138.154	271.409	4	4	2
7	152	309	194.61	474.747	4	5	6
8	70	70	173.782	259.622	1	2	4
9	70	70	78.0122	105.154	4	0	0
10	291	580	378.99	797.838	6	10	6
11	719	1338	884.767	1740.22	18	17	20
12	469	926	537.345	1087.02	9	12	8
13	160	224	189.024	283.007	5	2	3
14	257	377	394.191	861.394	11	8	17
15	39	39	61.3848	143.154	1	2	3
16	126	283	261.392	474.943	3	7	5
17	167	329	206.51	643.529	4	4	9
18	580	904	697.612	1291.34	11	16	14
19	255	480	318.907	643.872	6	8	7
20	306	518	353.25	648.63	5	8	9
21	499	821	591.002	981.436	13	17	14
22	168	295	232.551	427.333	6	6	7
23	153	207	194.782	394.978	5	4	6
24	370	622	476.747	999.394	10	12	9

548_hyp_HP_z14c1-2: distributions, pre-validation, px